

6. Re-enter S/C shim currents from last iteration.
7. Perform another LVShim scan, analyze the results, and perform another passive shim iteration if analysis does not meet specifications.

3-7 Gradient LVShim

Overview

The goal of Gradient LVShim is to clean up the shim over a 22-cm DSV using gradients only.

Procedure

1. Perform a phantom setup, unless it was already done during Rough or Main LVShim. Refer to Section 3-3, *Phantom Setup*.
2. Perform LVShim using the following parameters:
 - Use a bandwidth of 200 Hz during scan acquisition.
 - Use manual prescan to locate and center the best possible signal.
 - Adjust the TG during manual prescan to optimize the signal.
 - Be sure to open the **Frequency** menu at the top of Manual Prescan window and select **[Save Frequency]** before scanning.
 - Refer to Section 3-4-1, *LVShim Scan*, for scan protocol.
3. Perform LVShim Analysis using the following:
 - Select **Gradient** as the Shim Type.
 - Choose the calibration file appropriate for the type of gradient coil you have. For all magnets except 0.5T, use the **_22dsv**. file for the epoxy-filled body coil.
 - Refer to Section 3-4-2, *LVShim Analysis*, for screen details.
4. Continue to scan and analyze until the delta currents for all three gradients are <1.000. This should take only a couple of iterations.
5. Check harmonic coefficients of X3 (3,-3), Y3 (3,3), Z3 (3,0), Z2Y (3,1) and Z2X (3,-1). If changing the bandwidth from 500 to 200 Hz dramatically changed these harmonics, then check your Grafidy data at this point. If you have to redo the Grafidy, please check your LVShim results again.
6. Delete the LVShim images when finished shimming in order to maximize image disk space for the customer.
7. For the **TwinSpeed** system, repeat the calibration for the Zoom coil.

3-8 Troubleshooting and Solutions

3-8-1 Localizer Scan

Overview

The main purpose of the Localizer Scan is to check the landmark and verify that the ROI is in the center of the image.

Tools Required:

- LVShim Phantom Assembly, 2125245
- Nesting Plate Assembly, 2125247

Note

Foreign material (such as staples, metal filings, dirt, etc.) on the LVShim phantom or nesting plate will alter shim harmonics, ensuring that the system will not properly shim. Therefore, before using the phantom and nesting plate, be sure that all foreign material has been cleaned off.

Procedure

1. Refer to Section 3-3, *Phantom Setup*, if the phantom has not been previously set up.
2. At the operator workspace, select the Scan icon in the desktop control panel, if you have not already done so.
3. Click on **[Autoview]**, just below the Autoview image display screen. Your images will be displayed automatically.
3. Set up the Localizer scan as follows:
 - a. In Patient Register, click **[New Pt]**.
 - b. In Patient Information, enter:
Patient ID **geservice** and press **<Enter>**
Patient Name **LVShim** and press **<Enter>**
Weight (lb.) **300**
Click **[Landmark]**
Next to the Landmark box, click **[>]** and select **[Sternal Notch]**.
 - c. At the keypad on the front magnet enclosure, press LANDMARK and MOVE TO SCAN to position the landmarked phantom at the center of the magnet bore.
 - d. At the operator workspace, prepare the system for LVShim scan using the "Service Protocols" procedure located on the service methods CD-ROM, or, for the alternate proprietary procedure, see below.

Note

Perform the DC offset (from Diagnostics menu) before **each** LVShim scan.

This alternate proprietary procedure is available for GE use, and to sites with a valid Advanced Service Package Limited License.

In the Protocol field, type **o.19.1** (o = Other, 19 for protocol, 1 = series number).

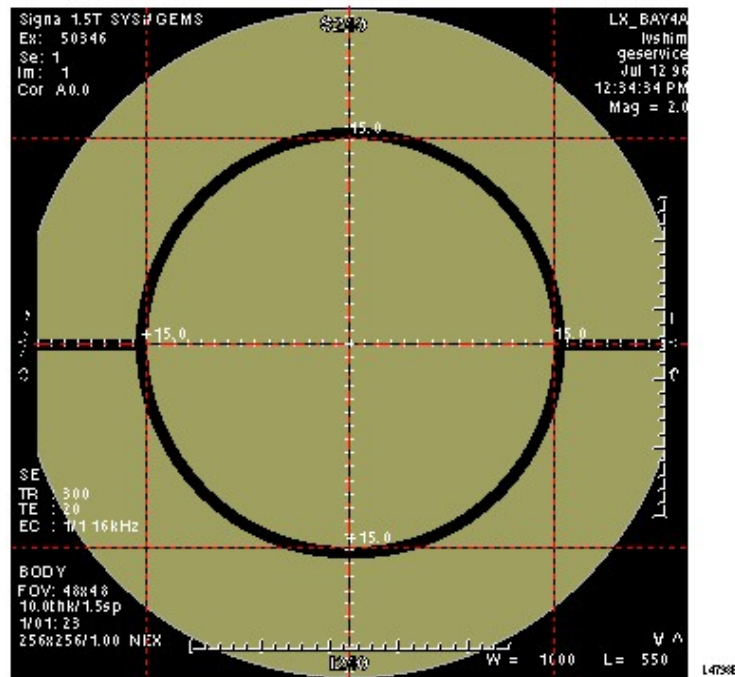
For **TwinSpeed**, ensure **GradMode** is set to **WHOLE**. There is no need to repeat this for **ZOOM**.

- e. Click **[Save Series]**.
- 4. Click **[Scan]**.

Note

If you are performing Rough LVShim, you may have to use [Manual Prescan] to find the center frequency. Use [Center Freq Coarse (CFL)] to center on the frequency peak as necessary. Be sure to save the center frequency value before exiting manual prescan. Select the Frequency menu at the top of Manual Prescan window and select **[Save Frequency]**. You should also write down the center frequency value for future reference.

- 5. When the image is displayed in the Autoview window, select the Display icon in the desktop control panel. To display the image, select the Exam, Series, and Image, and click **[Viewer]**. Do the following to verify that the phantom is centered in the Z direction:
 - a. Click **[User Prefs]**.
 - b. In the User Preferences pop-up window, find **Grid Prefs** and click **[Customize...]**.
 - c. Change Grid Spacing (mm) to 150, click **[OK]**, and then click **[Apply]**.
 - d. Select the button with the grid symbol to turn on the grid. See Illustration 3-6, Centering LV Phantom, for an example of an image with the alignment grid on.



CENTERING LVSHIM PHANTOM
ILLUSTRATION 3-6

- e. Adjust Window (W) and Level (L) settings until the edges of the phantom image can be clearly seen.
- f. Verify that the inner phantom sphere is centered within the 150-mm grid in the S/I direction, as shown in Illustration 3-6. If the phantom is not centered, move the phantom in or out of the bore as necessary (based on the grid markings), then landmark again and re-scan.

3-8-2 Crude Image Procedure

Overview

The Crude Image procedure is usually used when there is no signal or a very weak signal (such as troubleshooting for reversed I and Q cable).

Tools Required

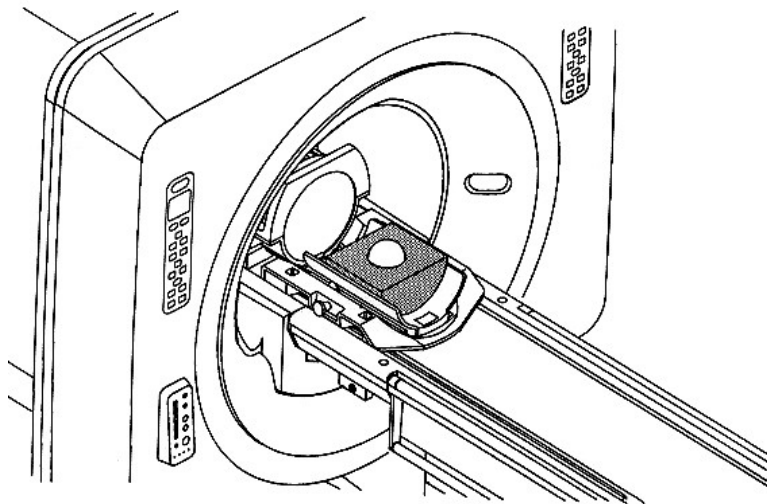
- 100mm Sphere phantom, 46-317586G1
- EPI Foam Positioner, 2138870

Note

Foreign material (such as staples, metal filings, dirt, etc.) on the LVShim phantom or nesting plate will alter shim harmonics, ensuring that the system will not properly shim. Therefore, before using the phantom and nesting plate, be sure that all foreign material has been cleaned off.

Procedure

1. Set up the system to perform a head coil scan. Place the EPI foam phantom holder and 100-mm sphere in the head coil as shown in Illustration 3-7.



EPI PHANTOM POSITIONING IN HEAD COIL
ILLUSTRATION 3-7

2. Landmark on the center of the head coil, adjust the positioner / sphere to axial alignment light, and press the MOVE TO SCAN button.
3. At the operator workspace, select the Scan icon in the desktop control panel, if you have not already done so.
4. Click on **[Autoview]**, just below the Autoview image display screen. Your images will be displayed automatically.
5. Set up the scan as follows:
 - a. In Patient Register, click **[New Pt]**.
 - b. Enter Patient Information:
Patient ID **geservice**
Patient Name **LVShim**
Weight (lb.) **100**
Click **[Landmark]**
Next to the Landmark box, click **[>]** and select **[Nasion]**.
 - c. At the keypad on the front magnet enclosure, press LANDMARK and MOVE TO SCAN to position the landmarked phantom at the center of the magnet bore.

- d. At the operator work space, prepare the system for LVShim scan using the "Service Protocols" procedure located on the service methods CD-ROM, or, for the alternate proprietary procedure, see below.

This alternate proprietary procedure is available for GE use, and to sites with a valid Advanced Service Package Limited License.

In the Protocol field, type **o.19.1** (o = Other, 1 = series number).

For **TwinSpeed**, ensure **GradMode** is set to **WHOLE**. There is no need to repeat this for **ZOOM**.

- e. Make the following changes to the protocol:

Patient Position Coil: **HEAD**.

Acquisition Timing Phase: **128**, select **Autoshim**

Scanning Range FOV: **12**

- f. Click [**Save Series**].

6. Click [**Manual Prescan**]. While in the [**Center Freq Fine (CFH)**] mode, verify that the frequency peak is centered in the display. Center the frequency peak as necessary. After locating the best signal, open the **Frequency** menu at the top of Manual Prescan window and select [**Save Frequency**] to save the new center frequency. If a signal cannot be found, measure the center frequency at the center of the bore using a magnetometer probe. Repeat this step and enter probe measurement for center frequency. If a signal still cannot be found then try re-entering the shim currents.
7. Click [**Done**] to exit manual prescan.
8. Click [**Auto Prescan**] to perform Autoshim. When Auto Prescan is complete, click [**Auto Prescan**] again. It is necessary to perform a second Auto Prescan to get accurate center frequency and Gradshim values. Auto Prescan performs a frequency adjustment prior to the Autoshim adjustment and there is interaction between the two processes.
9. Click [**Manual Prescan**]. Verify the Gradshim values and the System Frequency, and verify that the waveform is centered. Then open the **Frequency** menu at the top of Manual Prescan window and select [**Save Frequency**] to save the new center frequency.
10. Click [**Scan**].
11. When the scan is completed, view the image.
 - a. If the image resembles the phantom, continue to Section 3-8-3, Axial Currents.
 - b. If the image does not resemble the phantom, and any additional adjustments made by Autoshim do not affect the image, then perform the mechanical plotting procedure in the appropriate magnet service manual.

3-8-3 Axial Currents

Development of LVShim uncovered some differences in the way certain magnets behave, especially with respect to even axial harmonics (primarily Z4 and Z6). With some magnets (especially the Active Shield types S-IV, S-V, S-X, S-XC_1 and S-XC_2), it is difficult to reduce both the Z4 and Z6 harmonics (they tend to affect each other significantly). The LVShim analysis software may request the entry of currents as high as 20 amps into the Z4 and Z6 coils. The solution to this situation is to run LVShim and use no Z6 coil option. This allows the software to solve for the Z4 harmonic without calculating excessively high currents. This also prevents the Z6 coil from pushing the magnet center frequency outside of the specification. The CX and LCC magnets should use the Z6 coil.

3-8-4 Possible Need to Repark a Magnet

There is a possibility, after shimming a magnet on a new install, that the center frequency of the magnet could fall outside the allowable bandwidth range for the system. Therefore, it will be necessary after shimming to check the magnet center frequency and re-park the magnet. Refer to Table 3-2 for center frequency specifications.

TABLE 3-2
MAGNET FINAL PARKING FREQUENCY SPECIFICATIONS

MAGNET TYPE	MAGNET FIELD STRENGTH GAUSS (FREQUENCY)	BANDWIDTH GAUSS (FREQUENCY)
1.5T	15,000 G (63,864000 MHz)	+/- 7.5 G (+/- 31.932 kHz)
1.0T	10,025 G (42,682440 MHz)	+/- 5.0 G (+/- 21.288 kHz)

If the system cannot meet the LVShim Specifications during Main LVShim, and requires more than 20 amps on the axial shim coils, please contact the On-Line Center for help.

Signa Horizon systems with a GRAM may cause stability issues with the LVShim PSD, which may cause some harmonics to diverge (especially X3 and Y3). If you are shimming a system with a GRAM, and you notice certain harmonics diverging, use the half-current shim calibration files to force the convergence of these harmonics.

3-8-5 Gradient Shim

LVShim can usually shim a magnet starting from magnet ATR currents, eliminating the need for any mechanical plotting; however, there are a few situations that may make this goal somewhat difficult to achieve. Gradient Shim is a very useful tool at this point.

The system must be able to achieve at least some images in the center for LVShim to work. The image can be very distorted and weak, but you must have something to start with. Usually the magnet ATR currents are enough to give you a useable first image. However, the signal may be very weak and hard to find due to different site conditions.

It is very important that you use Manual Prescan to find and center on the best possible signal. You may have to increase the Receive Gains (R_1 and R_2) during manual prescan to find the signal. Also, you should adjust the TG until the signal is maximized. After locating the best signal, be sure to open the **Frequency** menu at the top of Manual Prescan window and select **[Save Frequency]** to save the new center frequency.

If the image is too weak or distorted for LVShim to analyze, you may be able to use the gradients to "stretch" the image into shape. This will take some experimentation, but it may help you avoid mechanical plotting. After the magnet is rough shimmed, the gradients should be set back to zero and their effects translated to shim currents before performing Main Shim.

When the gradients are set back to zero, an equivalent current should be entered into the X, Y, and Z1 shim power supplies before the next iteration. Refer to Table 3-3 for the multiplier to use when converting the gradient units into shim power supply currents. For example, for an S-IV magnet with an epoxy-filled gradient coil, if 280 units of Z gradient were used during rough shimming, the equivalent amount of shim current to enter into the Z-shim power supply would be 1.288 amps ($.0046 \times 280$). As shown in Table 3-3, for all magnets except Oxford, two shim coils (T2-2 and T2-4) must be used to get an effect equivalent to the X gradient, and two shim coils (T1-2 and T1-4) must be used to get an effect equivalent to the Y gradient.

Note

If the gradients were used to "stretch" the image during Rough LVShim, they should be set back to zero before you perform Main LVShim.

TABLE 3-3
GRADIENT TO SHIM COIL EQUIVALENTS

MAGNET TYPE (Note 1)	GRADIENT	SHIM COIL	MULTIPLIER FOR BRM (Note 2)	MULTIPLIER FOR CRM (Note 2)
S-II, S-III, S-IV, S-V, S-X, S-XC_1, S-XC_2, Cx 1.5T, Cx 1.0T	Z	AX1	+0.0042	TBD
	Y	T1-2	+0.0080	
		T1-4	+0.0080	
	X	T2-2	-0.0077	
		T2-4	-0.0077	
Note 1: No equivalents are given for Oxford and S-I Magnets. Rough Shim will not be necessary for Oxford and S-I Magnets, since they have already been previously shimmed.				
Note 2: Multiplier value is to convert Gradshim value to shim coil current.				

This table requires updating for the TwinSpeed.

3-8-6 Additional Passive Shimming Required

For S-IV, S-X, S-V, S-XC_1, and S-XC_2 Magnets, if additional passive shimming is required beyond the passive shimming performed in Florence (usually due to site environmental conditions), you will need to perform a passive shim. The need for additional passive shimming at the site is minimized because Florence now shims Active Shield magnets using 45-cm spherical plotting during acceptance testing.